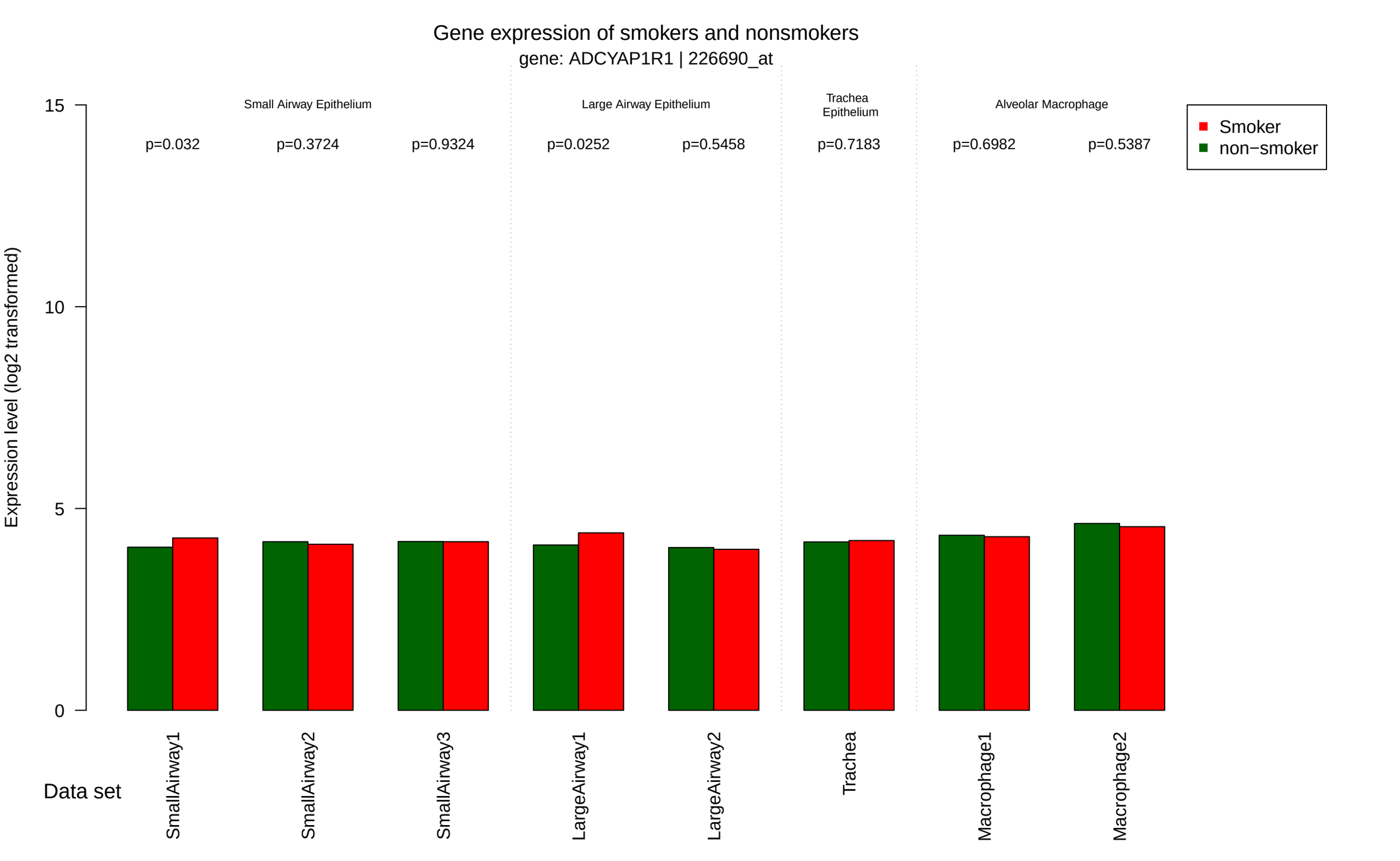
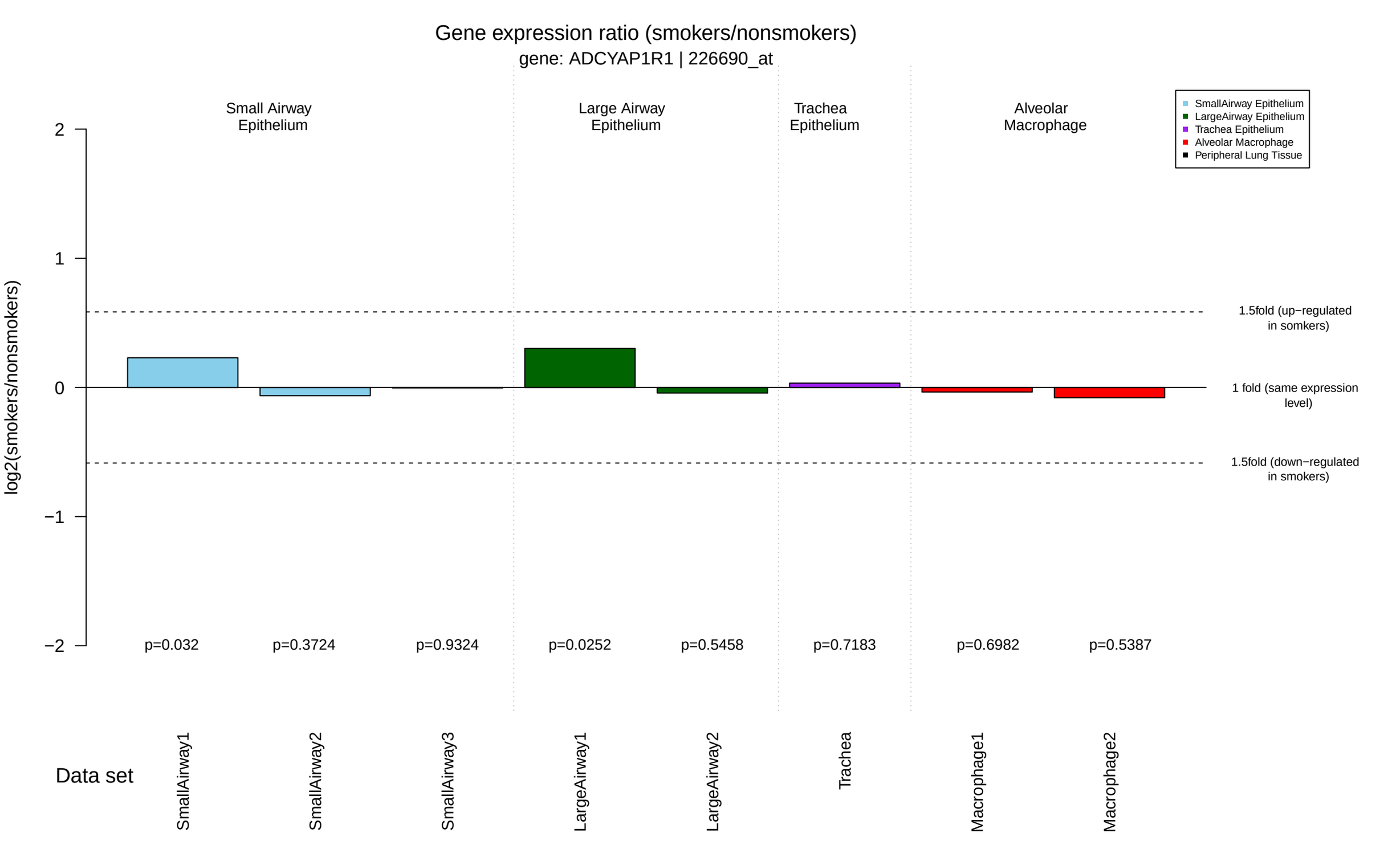
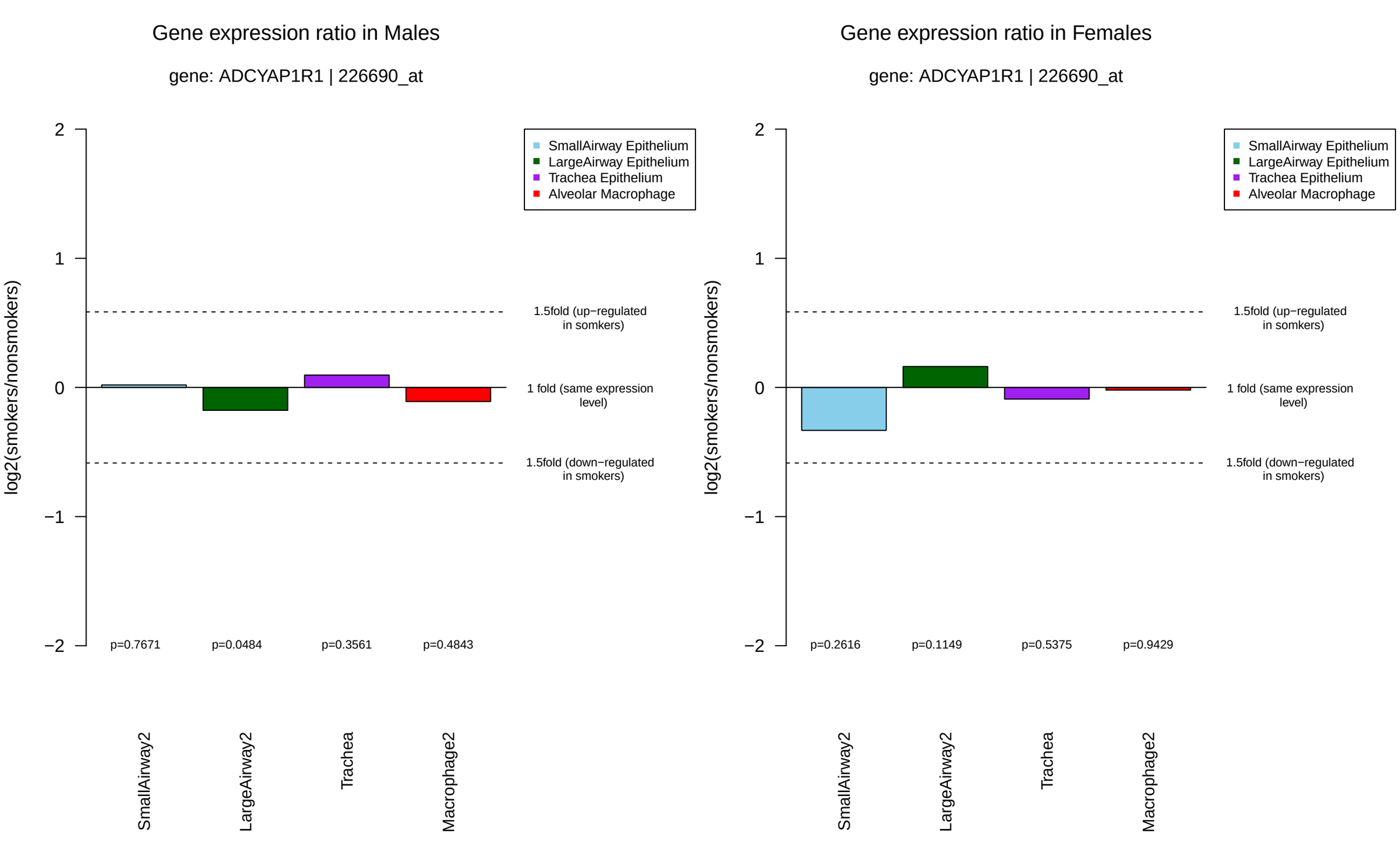
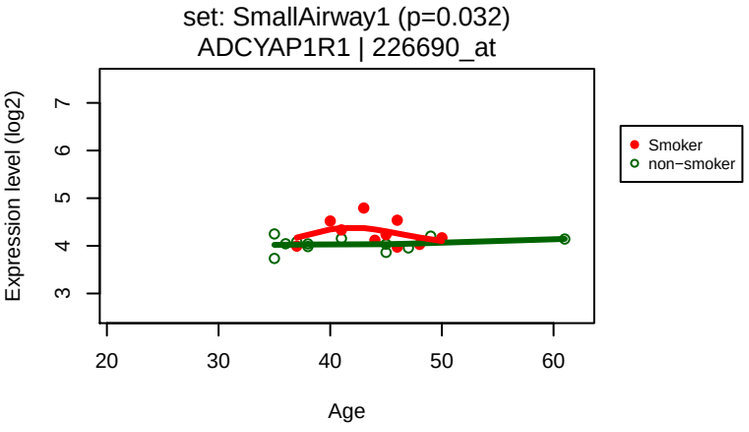
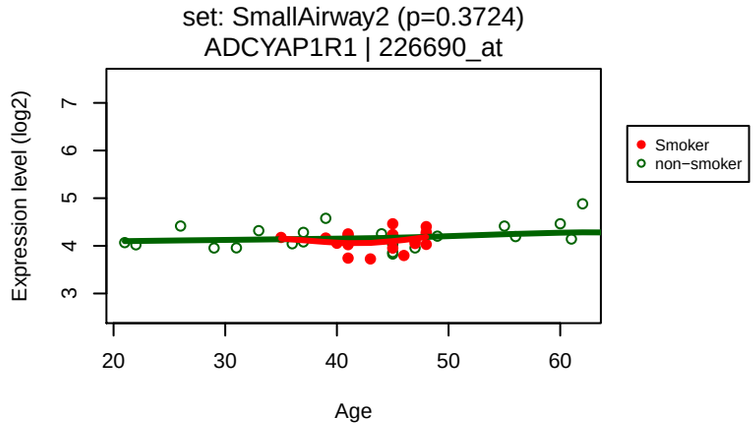
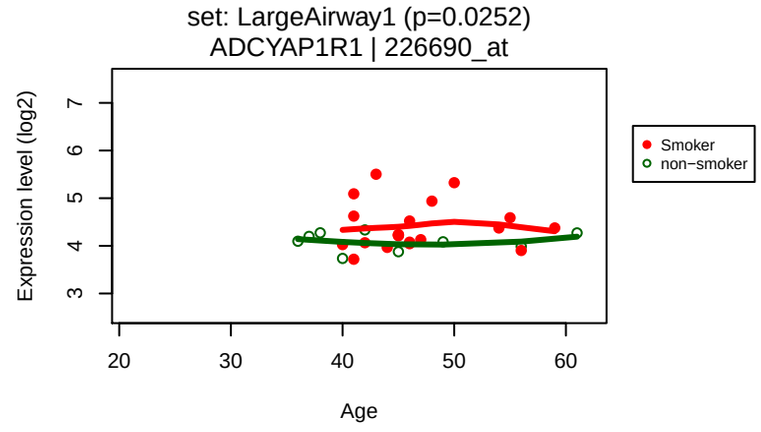

Gene expression of smokers and nonsmokers
gene: ADCYAP1R1 | 226690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.032
p=0.3724
p=0.9324
p=0.0252
p=0.5458
p=0.7183
p=0.6982
p=0.5387
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

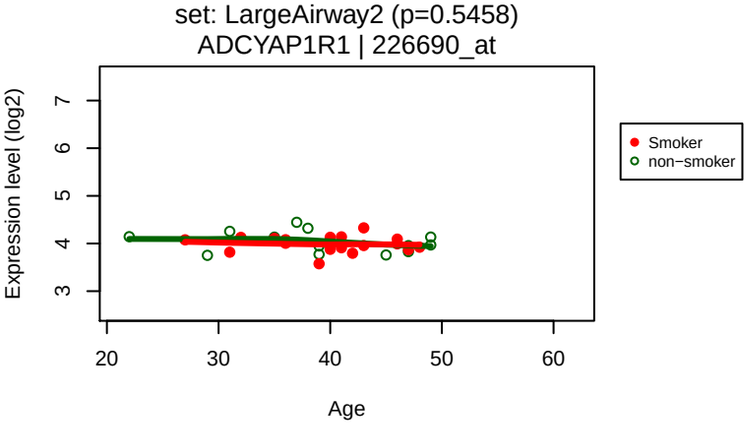

Gene expression ratio (smokers/nonsmokers)
gene: ADCYAP1R1 | 226690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.032
p=0.3724
p=0.9324
p=0.0252
p=0.5458
p=0.7183
p=0.6982
p=0.5387
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

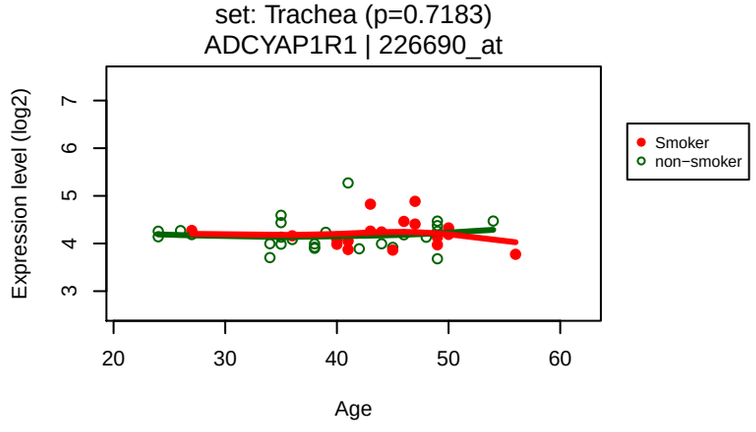

Gene expression ratio in Males
gene: ADCYAP1R1 | 226690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7671
p=0.0484
p=0.3561
p=0.4843
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCYAP1R1 | 226690_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2616
p=0.1149
p=0.5375
p=0.9429
SmallAirway2
LargeAirway2
Trachea
Macrophage2

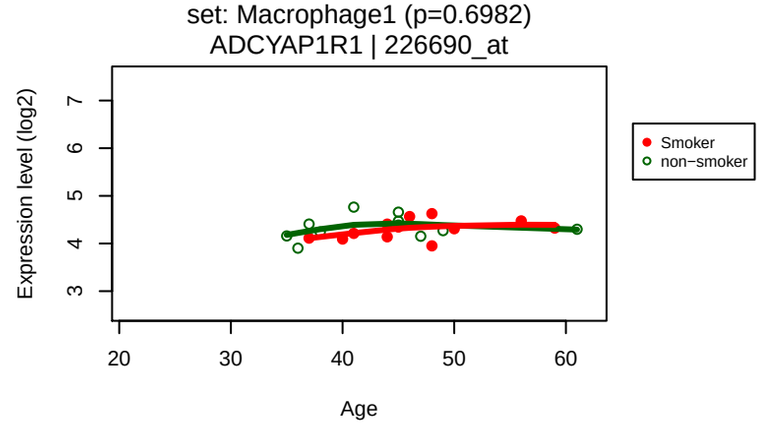

set: SmallAirway1 (p=0.032)
ADCYAP1R1 | 226690_at
Expression level (log2)
Age
Smoker
non−smoker

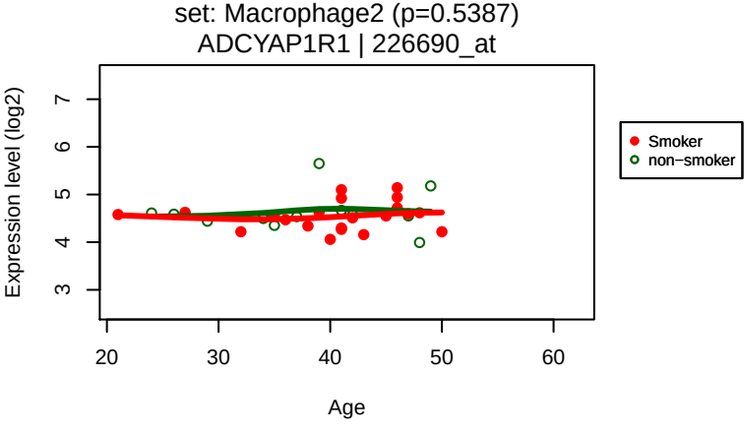

set: SmallAirway2 (p=0.3724)
ADCYAP1R1 | 226690_at
Expression level (log2)
Age
Smoker
non−smoker

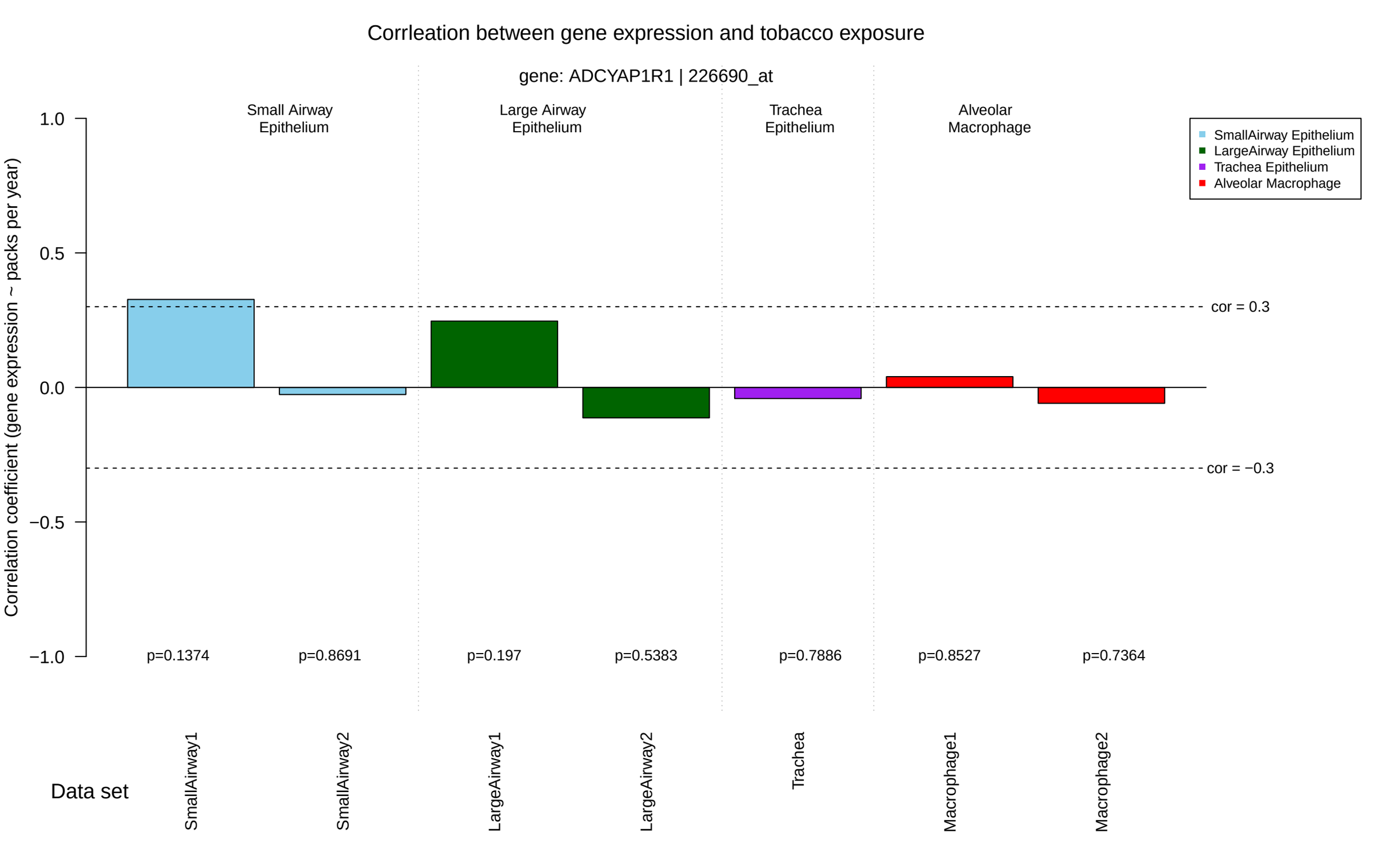

set: LargeAirway1 (p=0.0252)
ADCYAP1R1 | 226690_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5458)
ADCYAP1R1 | 226690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7183)
ADCYAP1R1 | 226690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6982)
ADCYAP1R1 | 226690_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5387)
ADCYAP1R1 | 226690_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCYAP1R1 | 226690_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1374
p=0.8691
p=0.197
p=0.5383
p=0.7886
p=0.8527
p=0.7364
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2